Santhosh

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EDUCATION

MS in Electrical and Electronics Engineering (Embedded Systems and IoT) | University of Colorado at Boulder **May 2023 - *GPA: 3.88***
Coursework: *Embedded System Design, Advanced Embedded Software Development, Compiler Construction, Concurrent Programming*

B.E. in Electronics and Communication Engineering | *Visvesvaraya Technological University, India* ***May 2021 - GPA: 8.8/10***

TECHNICAL SKILLS

- **Programming Languages**: C | CPP | Python | x86 assembly
- **Technologies:** Embedded Linux | Embedded Systems and Software | Firmware | IoT | RTOS | Robotics | Device drivers | Shell scripting | Simics | DML | OpenMP | SQLite | ORM | swupdate | NGINX
- **Hardware:** Beagle bone Black & Blue | Raspberry Pi | MSP 432 | Jetson Nano | FRDM-KL25Z | LMX2572P | STM, Atmel, Arm, and TI (Sitara) controllers and processors | ESP8266 | Intel Server Processors
- **Communication Protocols:** UART | SPI | I2C | RS232 | LoRa | Bluetooth LE| Wi-Fi | HTTP| MQTT | PCIe | CXL | DMA
- **Software/Tools:** GCC | GDB | Valgrind | TI CCS | MATLAB | Docker | Git | Busy box | Yocto | Crosstool-ng | Jenkins

EXPERIENCE HIGHLIGHTS

Hach Company, Loveland, CO, June 2023 - Present
Embedded Software / Firmware Engineer

- Play an integral role in the Platform Development Team, focusing on core functionalities that support three concurrent new product development teams.
- Lead the design and implementation of the data acquisition and management process on the controller, including efficient data handling via HTTP, MQTT, SQLite databases, Object Relational Mapping (ORM), and Object-Oriented Programming (OOP).
- Actively involved in the development of a new controller based on the platform, contributing to its architecture and integration.
- Oversee the firmware update process integrated across three products, ensuring seamless updates and compatibility within the platform.
- Develop and maintain BSP components (e.g., Device Trees), package management, and Protocol Buffers for common device Interface.
- Manage CI/CD pipelines and Jenkins workflows with Yocto environment to streamline development process.

INTEL CORPORATION, Santa Clara, CA, Summer & Fall 2022
System Software Engineer Intern

- Development and Testing of Software Simulator Models (virtual platform) for Server System on Chip and Platform.
- Collaborated in development and debugging of simulated transaction-level models for PCI-Express and CXL Links.
- Standalone Simics IP Services Driver Firmware model development for peripherals isolation testing.
- SoC PCIe root port enumeration, link training, bifurcation, memory interleaving software modeling and testing.
- PCIe 6.0 and CXL 3.0 devices Functional and Behavioral models development and testing.

UNIVERSITY OF COLORADO BOULDER, CO, JAN/ 2023 – Present
Graduate Teaching Assistant, Compiler Construction

- Built a compiler for python using python and C with various compiler optimization techniques.
- Compiler features include type checking, register allocation, liveness analysis, LVN (folding), Dead Store Eliminations, Control Flow Graphing, Grammar check, Escape Analysis.

JOINT INSTITUTE FOR LABORATORY ASTROPHYSICS, CU Boulder, CO, Aug/ 2021 – May/ 2022
Firmware Engineer

- Supported the development of Embedded Linux, Digital Systems, and Drivers, assisting in the management of end-to-system fimware development and design and implementation of solid platform and services.

PROJECTS

Self-Balancing Line Follower, An Assistant Bot, *Spring 2021*
Implemented a balancing, unstable equilibrium system to enable the robot to cope with temporary disturbance.
VIGIL, The Sky Sniper, *2019 to 2021*
Developed an autonomous drone using Beagle bone Black, LoRa to reduce latency in taking actions and data storage.
Morse Code Interpreter, *Fall 2021*
FRDM-KL25Z based More code interpreter, which converts the input code from the tap-button to light source output
BBB Video Streaming Device, *Spring 2022*
Built a custom Linux distribution for Beaglebone Black using Yocto, which uses GStreamer UDP pipelines to unicast and multicast videos across multiple devices.